Algorithm No : **2.2**

Algorithm Name: **Program of Quadratic Equation**

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define endl "\n"
#define newl cout<<endl
int main()
{
   float A,B,C;
   cin>>A>>B>>C;
   float D=B*B-4*A*C;
   if(D>0)
   {
      float x1=(-B+sqrt(D))/(2*A);
      float x2=(-B-sqrt(D))/(2*A);
      cout<<x1<<" "<<x2;
   }
   else if(D==0)
   {
      float x=-B/(2*A);
      cout<<"UNIQUE SOLUTION, "<<x;
   }
   else
   {
      cout<<"NO REAL SOLUTION";
   }
   newl;
}
```

Input:	**Output:**
5 4 3	NO REAL SOLUTION
1 -2 1	UNIQUE SOLUTION, 1
1 3 2	-1 -2

Algorithm No : **2.3**

Algorithm Name: **Program of Largest Element in an array**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define endl "\n"
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    int Arr[n+1];
    for(int i=1;i<=n;i++)
    {
        cin>>Arr[i];
    }
    int loc=1,maxx=Arr[1];
    for(int i=2;i<=n;i++)
    {
        if(maxx<Arr[i])
        {
            loc=i;
            maxx=Arr[i];
        }
    }
    cout<<loc<<" "<<maxx;
}
```

Input:	**Output:**
6 9 7 5 11 2 8	4 11
5 100 186 140 150 195	5 195

Algorithm No : **2.4**

Algorithm Name: **Program of Linear Search**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define endl "\n"
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    int Arr[n+1];
    for(int i=1;i<=n;i++)
    {
        cin>>Arr[i];
    }
    int a;
    cin>>a;
    int K=1,loc=0;
    while(K<=n)
    {
        if(Arr[K]==a)
        {
            loc=K;
        }
        K++;
    }
    loc!=0?cout<<loc:cout<<"NO VALUE";
    newl;
}
```

Input:	**Output:**
6 9 7 5 11 2 8 2	5
5 100 186 140 150 195 101	NO VALUE

Problem No: Practice Problem

Problem Name: **Largest and Second Largest**

Problem Link: **https://www.codechef.com/problems/LARGESECOND**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define P_B push_back
#define p_b pop_back
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define A_sort(x) sort(x.begin(),x.end())
#define D_sort(x) sort(x.rbegin(),x.rend())
#define endl "\n"
#define newl cout<<endl
int main()
{
   int t;
   cin>>t;
   while(t--)
   {
      int n;
      cin>>n;
      vector<int>A;
      map<int,int>mp;
      for(int i=0;i<n;i++)
      {
         int a;
         cin>>a;
         if(mp[a]==0)
         {
            A.P_B(a);
            mp[a]++;
         }
      }
      D_sort(A);
      cout<<A[0]+A[1]<<endl;
   }
}
```

Input:	**Output:**
2 3 4 1 6 7 3 7 2 1 1 5 3	10 12

Problem No: Practice Problem

Problem Name: **Four Tickets**

Problem Link: **https://www.codechef.com/problems/FOURTICKETS**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define P_B push_back
#define p_b pop_back
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define A_sort(x) sort(x.begin(),x.end())
#define D_sort(x) sort(x.rbegin(),x.rend())
#define yes_or_no cout<<"YES\n":cout<<"NO\n"
#define endl "\n"
#define newl cout<<endl
int main()
{
   int t;
   cin>>t;
   while(t--)
   {
        int t;
   cin>>t;
   while(t--)
   {
      int n;
      cin>>n;
      (n*4)<=1000?yes_or_no;
   }
   }
}
```

Input:	**Output:**
4 100 500 250 1000	YES NO YES NO

Problem No: Practice Problem

Problem Name: **Chef And Battery**

Problem Link: **https://www.codechef.com/problems/FIFTYPE**

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define mod0(x,y) x%y==0
#define endl "\n"
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    while(t--)
    {
        int n;
        cin>>n;
        if(ev(n)&&n<50)
        {
            cout<<(50-n)/2;
        }
        else if(od(n)&&n<50)
        {
            cout<<1+(53-n)/2;
        }
        else if(n>50)
        {
            if(mod0((n-50),3))
            {
                cout<<(n-50)/3;
            }
            else if((n-50)%3==1)
            {
                cout<<1+((n-50)+2)/3;
            }
            else
            {
                cout<<2+((n-50)+4)/3;
            }
        }
        else
        {
            cout<<0;
        }
        newl;
    }
}
```

Input:	**Output:**
4 51 50 23 0	2 0 16 25